MIDSEM

1 (a) $x = r\cos\theta \sec\phi$; $y = r\sin\theta \sec\phi$; $z = r\tan\phi$.

WRONG QUESTION

$$\frac{\partial x}{\partial r} = \cos\theta \sec\phi \; ; \; \frac{\partial y}{\partial r} = \sin\theta \sec\phi \; ; \; \frac{\partial z}{\partial r} = \tan\phi$$

$$\therefore \frac{\partial x}{\partial \theta} = -r\sec\phi \sin\theta \; ; \; \frac{\partial y}{\partial \theta} = r\sec\phi \cos\theta \; ; \; \frac{\partial z}{\partial \theta} = 0$$

$$\therefore \frac{\partial x}{\partial \phi} = r\cos\theta \sec\phi \tan\phi \; ; \; \frac{\partial y}{\partial \phi} = r\sin\theta \sec\phi \tan\phi \; ; \; \frac{\partial z}{\partial \phi} = r\sec^2\phi$$

Scale factors :

X

$$h_r = \sqrt{\left(\frac{\partial x}{\partial r}\right)^2 + \left(\frac{\partial y}{\partial r}\right)^2 + \left(\frac{\partial z}{\partial r}\right)^2}$$

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2 (a) $f(z) = u(x,y) + i\,v(x,y)$

For $f(z)$ to be analytic ; CR condition has to be satisfied :

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \; ; \; \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\therefore u = x^2 + xy - y^2 \quad \text{— (i)}$$

$$\therefore \frac{\partial u}{\partial x} = 2x + y = \frac{\partial v}{\partial y} \Rightarrow \boxed{v(x,y) = 2xy + \frac{y^2}{2} + f(x) \cancel{f(y)} + c}$$

$$\frac{\partial u}{\partial y} = -2y + x = -\frac{\partial v}{\partial x} \Rightarrow \boxed{v(x,y) = +2xy - \frac{x^2}{2} + c + g(y)} \quad \text{— (ii)}$$

Comparing (i) & (ii) :

$$f(x) = -\frac{x^2}{2} \; ; \; g(y) = \frac{y^2}{2}$$

$$\therefore \boxed{v(x,y) = -\frac{x^2}{2} + 2xy + \frac{y^2}{2} + c} \quad \text{(Ans)}$$

c — constant

(b) $f_1(z)$: $u = |y|$; $v = -x$

$$\therefore \frac{\partial u}{\partial x} = 0 = \frac{\partial v}{\partial y} \; ; \; \frac{\partial u}{\partial y} = \pm 1 \; ; \; \frac{\partial v}{\partial x} = -1.$$

For CR condition to be satisfied

(2)

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$\therefore \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ only when $\frac{\partial u}{\partial y} = 1 \Rightarrow \boxed{y > 0}$

$f_2(z)$: $u = |x|$; $v = |y|$.

$\therefore \frac{\partial u}{\partial x} = \pm 1 \neq \frac{\partial v}{\partial y}$; $\frac{\partial u}{\partial y} = 0 = \frac{\partial v}{\partial x}$

CR condition is satisfied both in when

$(x, y) > 0$ & $(x, y) < 0$

For both $f_1(z)$ & $f_2(z)$ to be analytic, $\boxed{(x, y) > 0}$

They are analytic only in the 1st quadrant.

(3) $f(z) = ze^z$

$= (x+iy)\, e^{x+iy}$

$= (x+iy)\, e^x \cdot e^{iy}$

$= (x+iy)\, e^x\, [\cos y + i \sin y]$

$\cancel{= e^x [x\cos y - y \sin y] +}$

$$\Rightarrow \boxed{f(x,y) = e^x [(x\cos y - y\sin y) + i(y\cos y + x\sin y)]} \quad \text{— (1)}$$

$\therefore Re[f(z)] = e^x [x\cos y - y\sin y] = u(x, y)$ — (a)

$\therefore Im[f(z)] = e^x [y\cos y + x\sin y] = v(x, y)$ — (b)

$\therefore \bar{\nabla}(Re[f(z)]) = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j}\right) \boxed{Re[f(z)]} \rightarrow u(x, y)$

$\cancel{= [e^x((x+1)(x\cos y - y\sin y) + \cos y)]\hat{i}}$

$\therefore \cancel{\partial Re[f(z)]}\ \frac{\partial u(x, y)}{\partial x} = e^x(x\cos y - y\sin y) + e^x \cos y$

$$\boxed{\frac{\partial u}{\partial x} = e^x[(x+1)\cos y - y\sin y]} \quad \text{— (ii)}$$

$\therefore \frac{\partial u}{\partial y} = -e^x x \sin y - e^x(\sin y + y\cos y)$

$$\Rightarrow \boxed{\frac{\partial u}{\partial y} = -e^x[(x+1)\sin y + y\cos y]} \quad \text{— (iii)}$$

$$\therefore \boxed{\bar{\nabla} u(x,y) = \text{(ii)}\ \hat{i} + \text{(iii)}\ \hat{j}} \quad - \text{(c)}$$

$$\therefore \frac{\partial v(x,y)}{\partial x} = e^x [\, y \cos y + x \sin y + \sin y \,]$$

$$\boxed{\frac{\partial v}{\partial x} = e^x [\, y \cos y + (x+1) \sin y \,]} \quad - \text{(iv)}$$

$$\frac{\partial v}{\partial y} = e^x [\, \cos y - y \sin y + x \cos y \,].$$

$$\boxed{\frac{\partial v}{\partial y} = e^x [\, (x+1) \cos y - y \sin y \,]} \quad - \text{(v)}.$$

$$\boxed{\bar{\nabla} v = \text{(iv)}\ \hat{i} + \text{(v)}\ \hat{j}} \quad - \text{(d)}$$

So @ $z_0 = 1+i$; $x_0 = 1$; $y_0 = 1$. Substituting the values into (c) & (d) :

$$\therefore \bar{\nabla} u|_{z_0} = e\,(\underbrace{2\cos(1) - \sin(1)}_{P})\,\hat{i} + e\,(\underbrace{2\sin(1) + \cos(1)}_{Q})\,\hat{j}$$

$$\therefore \bar{\nabla} v|_{z_0} = e\,(\underbrace{2\sin(1) + \cos(1)}_{Q})\,\hat{i} + e\,(\underbrace{2\cos(1) - \sin(1)}_{P})\,\hat{j}$$

$\bar{\nabla} u$ & $\bar{\nabla} v$ gives us the normal to these curves. If the normal to these curves @ z_0 is 1, then tangents are also 1.

$$\therefore \bar{\nabla} u|_{z_0} \cdot \bar{\nabla} v|_{z_0} = e^2 [\, PQ - PQ \,] = 0$$

$$\cancel{= e^2 [\, 4\cos(1)\sin(1) + 2\cos^2(1) - 2\sin^2(1) - \sin(1)\cos(1) - 4\sin(1)\cos(1)}$$

Hence, the unit vectors ∥ to the curves are orthogonal to each other. The unit vectors are:

$$u(x,y)|_{z_0} = \pm \frac{1}{\sqrt{5}} [\, (2\sin(1) + \cos(1))\,\hat{i} + (2\cos(1) - \sin(1))\,\hat{j} \,]$$

(Ans)

$$v(x,y)|_{z_0} = \pm \frac{1}{\sqrt{5}} [\, (2\cos(1) - \sin(1))\,\hat{i} \pm (2\sin(1) + \cos(1))\,\hat{j} \,]$$

(Interchanging expressions for $\bar{\nabla} u|_{z_0}$ & $\bar{\nabla} v|_{z_0}$ as we found normals)

(Ans)

From (ii), (iii), (iv) & (v):

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ & $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ $\rightarrow$ CR condition is satisfied.

$\therefore$ (c) · (d) $=$ $\boxed{\bar{\nabla} u \cdot \bar{\nabla} v = 0}$

Hence, this is true for any general point z. (Proved)

(4) (a) $u = \ln x$; $v = \ln y$

$\Rightarrow \dot{u} = \frac{\dot{x}}{x}$; $\dot{v} = \frac{\dot{y}}{y}$ — (i)

$\dot{x} = x \ln y = x v \Rightarrow v = \frac{\dot{x}}{x}$ — (ii)

$\dot{y} = -y \ln x = -y u \Rightarrow u = -\frac{\dot{y}}{y}$

Comparing (i) & (ii)

$\dot{u} = v$ & $\dot{v} = -u$

Let us assume that differentiat^n is wrt t.

$\therefore \frac{du}{dt} = v$ & $\frac{dv}{dt} = -u$.

$$\therefore \frac{d}{dt}\begin{bmatrix} u \\ v \end{bmatrix} = \overbrace{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}}^{M} \begin{bmatrix} u \\ v \end{bmatrix} \Rightarrow \frac{dX}{dt} = MX$$

$$\Rightarrow \int_0^X \frac{dX}{X} = M\int_0^t dt$$

$$\Rightarrow \boxed{X(t) = Ce^{tM}}$$

$$\Rightarrow \begin{bmatrix} u \\ v \end{bmatrix}_t = X$$

From the given initial conditions:

$x(0) = e^2 \Rightarrow u(0) = 2$

$y(0) = \frac{1}{e} \Rightarrow v(0) = -1$

$C = \begin{bmatrix} u(0) \\ v(0) \end{bmatrix}$

$$\therefore \boxed{\begin{bmatrix} u \\ v \end{bmatrix}_t = e^{tM} \begin{bmatrix} 2 \\ -1 \end{bmatrix}} \text{ — (iii)}$$

$$M^2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$$

$\therefore M^3 = M^2 M = -M$; $M^5 = M^3 M^2 = M$.

$\therefore M^4 = (M^2)^2 = I$.

The powers of M behave like powers of i.

(5)

$$\therefore e^{Mt} = I + Mt + \frac{(Mt)^2}{2!} + \frac{(Mt)^3}{3!} + \dots$$

$$= I + Mt + -\frac{It^2}{2!} - \frac{Mt^3}{3!} + \dots$$

$$= I\left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \dots\right)$$

$$+ M\left(t - \frac{t^3}{3!} + \frac{t^5}{5!} - \dots\right)$$

$$\boxed{e^{Mt} = I\cos t + M\sin t} \quad \text{— (iv)}$$

$$\therefore e^{Mt} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\cos t + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\sin t$$

$$e^{Mt} = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$$

From (iii):

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 2\cos t - \sin t \\ -2\sin t - \cos t \end{bmatrix}$$

$$\Rightarrow \quad u = 2\cos t - \sin t$$

$$v = -(2\sin t + \cos t)$$

$$\therefore u = \ln x \Rightarrow \boxed{x = e^u = e^{2\cos t - \sin t}}$$

$$\therefore v = \ln y \Rightarrow \boxed{y = e^v = e^{-(2\sin t + \cos t)}}$$

[Ans]

(b) $(D - a)x = f(t)$

$$\Rightarrow \frac{dx}{dt} - ax = f(t) \quad \text{— (i)}$$

For a linear differential equat^n of the form:

$$\frac{dx}{dt} + P(t)x = Q(t) \quad \text{— (ii)}$$

Integrating factor: $\boxed{I(t) = e^{\int P\,dt}}$ — (iii)

Multiplying (ii) with (iii):

$$\frac{dx}{dt} e^{\int P\,dt} + P(t)x\,e^{\int P\,dt} = e^{\int P\,dt} Q(t)$$

$$\Rightarrow \frac{d}{dt}\left[\left(e^{\int P\,dt}\right) x\right] = Q(t)\, I(t)$$

$$\Rightarrow \frac{d}{dt}\left(I(t)\, x\right) = Q(t)\, I(t)$$

$$\Rightarrow I(t)\, x = \int_0^t Q(t')\, I(t')\, dt'$$

$$\Rightarrow \boxed{x(t) = \frac{1}{I(t)} \int_0^t Q(t')\, I(t')\, dt'} \quad \text{— (iv)}$$

Comparing (1) & (iv):

$$Q(t) = f(t) \quad ; \quad P(t) = -a$$

$$I(t) = e^{\int P\,dt} = e^{-at}$$

$$\boxed{x(t) = e^{at} \int_0^t f(t')\, e^{-at'}\, dt'}$$

(Ans)
(Proved)

(5) (a) $M = \frac{1}{5}\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$

Let's say that $M = UDU^{-1}$

$$\Rightarrow M^2 = (UDU^{-1})(UDU^{-1})$$

$$= UD^2U^{-1}.$$

$$\Rightarrow M^3 = M^2 M$$

$$= (UD^2U^{-1})(UDU^{-1})$$

$$= UD^3U^{-1}$$

$$\therefore \boxed{M^n = UD^nU^{-1}} \quad \text{— (1)}$$

Here D is a diagonal matrix of all the eigenvalues of M & U is a matrix containing all eigenvectors of M.

Eigenvalues:

Let λ be the eigenvalue of M.

$$\therefore |M - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} \frac{2}{5} - \lambda & \frac{1}{5} \\ \frac{3}{5} & \frac{4}{5} - \lambda \end{vmatrix} = 0 \Rightarrow \left(\frac{2}{5} - \lambda\right)\left(\frac{4}{5} - \lambda\right) - \frac{3}{25} = 0$$

$$\Rightarrow 25\lambda^2 - 30\lambda + 5 = 0$$

$$\Rightarrow \boxed{5\lambda^2 - 6\lambda + 1 = 0}$$

$$\therefore \lambda = \frac{6 \pm \sqrt{36-20}}{10}$$

$$= \frac{3}{5} \pm \frac{2}{5}$$

$$\therefore \lambda_+ = 1 \; ; \quad \lambda_- = \frac{1}{5} \quad \text{— (ii)}$$

$$\therefore D = \begin{bmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{bmatrix} = \boxed{\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{5} \end{bmatrix} = D} \quad \text{(Ans)}$$

— (iii)

Eigenvectors :

$\lambda_+ = 1$:

Let the eigenvector be $A_+ = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$

$$\therefore (M - \lambda_+ I) A_+ = 0$$

$$\Rightarrow \begin{pmatrix} \frac{2}{5} - 1 & \frac{1}{5} \\ \frac{3}{5} & \frac{4}{5} - 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$\Rightarrow -\frac{3}{5} a_1 + a_2 = 0 \quad \Rightarrow \boxed{a_2 = 3a_1}$$

Considering $a_1 = 1$; $a_2 = 3$. $\boxed{A_+ = \begin{pmatrix} 1 \\ 3 \end{pmatrix}}$

— (iv)

$\lambda_- = \frac{1}{5}$:

Let the eigenvector be $A_- = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$

$$(M - \lambda_- I)(A_-) = 0$$

$$\Rightarrow \begin{pmatrix} \frac{2}{5} - \frac{1}{5} & \frac{1}{5} \\ \frac{3}{5} & \frac{4}{5} - \frac{1}{5} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$\Rightarrow a_1 + a_2 = 0 \quad \Rightarrow \boxed{a_1 = -a_2}$$

Considering $a_1 = 1$; $a_2 = -1$ $\boxed{A_- = \begin{pmatrix} 1 \\ -1 \end{pmatrix}}$

— (v)

$$U = (A_+ \; A_-) = \boxed{\begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix} = U} \quad \text{— (vi)}$$

$\therefore |U| = -4.$

$$\therefore U^{-1} = -\frac{1}{4}\begin{pmatrix} -1 & -1 \\ -3 & 1 \end{pmatrix} = \boxed{\frac{1}{4}\begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix} = U^{-1}} \quad \text{— (VII)}$$

From (III), (VI) & (VII):

$$\cancel{M = UDU^{-1}}$$

$$M = UDU^{-1} = +\frac{1}{4}\begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{5} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}$$

$$\therefore M^n = UD^nU^{-1} \quad [\text{from } (I)]$$

$$\therefore D^n = \begin{pmatrix} 1^n & 0 \\ 0 & \frac{1}{5}^n \end{pmatrix} \quad [\text{from } (III)]$$

$$\Rightarrow \cancel{UD^nU^{-1}}$$

$$\therefore UD^nU^{-1} = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 5^{-n} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}$$

$$= \frac{1}{4}\begin{pmatrix} 1 & 5^{-n} \\ 3 & -5^{-n} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}$$

$$= \boxed{\begin{pmatrix} 1 + 3\times 5^{-n} & 1 + 5^{-n} \\ 3(1-5^{-n}) & 3 - 5^{-n} \end{pmatrix}\frac{1}{4} = M^n} \quad \text{(Ans)}$$

(b) $\lim_{n\to\infty} M^n \longrightarrow$ all 5^{-n} terms vanish.

$$= \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix} \quad \text{(Ans)}$$

(I) (b) $(\bar{\nabla}\times\bar{A})_k = \epsilon_{ijk}\nabla_i A_j = \epsilon_{ijk}\partial_i A_j$

$$\therefore [\bar{A}\times(\bar{\nabla}\times\bar{A})]_i = \epsilon_{ilm} A_l (\bar{\nabla}\times\bar{A})_m$$

$$= \epsilon_{ilm} A_l \epsilon_{mab} \nabla_a A_b.$$

$\bar{A}\cdot\bar{B} = \delta_{ij} A_i B_j$

$\bar{\nabla}\cdot\bar{A} = \delta_{ij}\partial_i A_j$

$$= (\epsilon_{ilm}\epsilon_{abm}) A_l \partial_a A_b.$$

$$= (\delta_{ia}\delta_{lb} - \delta_{ib}\delta_{la}) A_l \partial_a A_b.$$

$$= \delta_{ia} A_l \partial_a A_b \delta_{lb} \text{ (X)} - \delta_{ib}\delta_{la} A_l \partial_a A_b$$

Don't combine this term as denoted, use delta contract^n

$$= \cancel{A_l \partial_i A_l - A_l \partial_l A_i \delta_{ib} A_b (\bar{\nabla}\cdot\bar{A})}$$

Now

$$\cancel{= \frac{1}{2}(2A_l \partial_i A_l) - A_i(\bar{\nabla} \times \bar{A})}$$

$$\cancel{= \frac{1}{2}[\partial_i (A^2)] - A_i(\bar{\nabla} \cdot \bar{A})}$$

$$= \frac{1}{2} \cancel{\delta_{ia} \partial_a (\underbrace{\bar{A} \cdot \bar{A}}_{A^2})} - \delta_{ib} A_b (\bar{\nabla} \cdot \bar{A})$$

$$\cancel{= \delta_{ia} \partial_a A^2 - A_i(\bar{\nabla} \cdot \bar{A})}$$

$$= \frac{1}{2}[\partial_i (A_l A_l)] - A_i(\bar{\nabla} \cdot \bar{A})$$

$$= \frac{1}{2}(\partial_i A^2) - A_i(\bar{\nabla} \cdot \bar{A})$$

$$= \left[\frac{1}{2} \nabla A^2 - \bar{A}(\bar{\nabla} \cdot \bar{A})\right]_i$$

$$\therefore \boxed{\bar{A} \times (\bar{\nabla} \times \bar{A}) = \frac{1}{2} \nabla A^2 - \bar{A}(\bar{\nabla} \cdot \bar{A})}$$

(Proved)
(Ans)